

Model	AlpacaFarm ($\uparrow$)	GSM ($\uparrow$)	ToxiGen ($\downarrow$)	TruthfulQA ($\uparrow$)	
	Win rate	Acc.	% Toxic	MC Acc.	% Info + True
<i>7B</i>					
Directly tuned	82.5	23.0	0.0	55.9	81.3
<i>13B</i>					
Base (untuned)	2.1	6.6	70.4	38.6	49.1
Proxy-tuned	83.4	26.4	0.1	57.4	82.0
Directly tuned	87.3	32.4	0.0	61.6	80.4
<i>70B</i>					
Base (untuned)	3.7	9.6	67.4	42.3	53.9
Proxy-tuned	88.0	32.0	0.0	59.2	85.1
Directly tuned	90.4	51.8	0.0	68.3	79.6

Table 2: **Results for instruction-tuning.** For each model size, **Base** refers to the pretrained LLAMA2 model, **Directly tuned** refers to LLAMA2-CHAT, and the **Proxy-tuned** model always uses LLAMA2-7B-CHAT as the expert and LLAMA2-7B as the anti-expert. Overall, proxy-tuning dramatically improves performance over the base model, on average closing 91.1% and 88.1% of the gap with the corresponding CHAT model at 13B and 70B size, respectively. It also outperforms the small expert alone in all scenarios except a 0.1% difference in ToxiGen.

$$p_{\tilde{\mathcal{M}}}(X_t | x_{<t}) = \text{softmax}[s_{\mathcal{M}}(X_t | x_{<t}) + s_{\mathcal{M}^+}(X_t | x_{<t}) - s_{\mathcal{M}^-}(X_t | x_{<t})] \quad (1)$$

Intuitively, Eq. (1) applies the result of tuning at a smaller scale (i.e., the learned difference between $\mathcal{M}^-$ and $\mathcal{M}^+$) to a larger base model ($\mathcal{M}$). Alternatively, by grouping Eq. (1) as $s_{\mathcal{M}^+} + (s_{\mathcal{M}} - s_{\mathcal{M}^-})$, we can also think of the ensemble as contrasting a large and small pretrained model in the style of contrastive decoding (Li et al., 2023), and applying the result to a small tuned model, thus giving the small expert the benefit of larger-scale pretraining.

The goal of proxy-tuning is to close the gap between the base model $\mathcal{M}$ and its directly tuned version, without modifying (or even accessing) the parameters of $\mathcal{M}$.

3 Instruction-Tuning Experiments

First, we evaluate proxy-tuning for instruction-tuning. We use the LLAMA2 model family (Touvron et al., 2023), which includes both BASE models pretrained on text, and CHAT models further aligned for dialogue by undergoing additional stages of supervised instruction-tuning and reinforcement learning from human feedback (RLHF; Stiennon et al., 2020). Both BASE and CHAT models have variants at 7B, 13B, and 70B parameters. We use 7B-CHAT as the expert $\mathcal{M}^+$ and 7B-BASE as the anti-expert $\mathcal{M}^-$, and steer 13B- and 70B-BASE as $\mathcal{M}$.

3.1 Datasets

We evaluate on **GSM** (Cobbe et al., 2021), a dataset of arithmetic word problems, **AlpacaFarm** (Dubois et al., 2023), which contains open-ended instructions, **Toxigen** (Hartvigsen et al., 2022), which evaluates toxicity of model generations, and **TruthfulQA** (Lin et al., 2022), which contains often misleading questions, and is evaluated in both a multiple-choice (MC) and open-ended question-answering setting. Please refer to Appendix A for details.

We use zero-shot prompting across all models, as we observe that LLAMA2-CHAT models struggle to follow the format of in-context examples. We use greedy decoding.

3.2 Results

Results are shown in Table 2. For **AlpacaFarm** and **GSM**, BASE models struggle to address the question; 70B-BASE achieves only 3.7% win rate on AlpacaFarm and 9.6% accuracy on GSM. Proxy-tuning 70B-BASE improves performance dramatically, to 88.0% on AlpacaFarm

and 32.0% on GSM. For AlpacaFarm, this is only 2.4% short of the CHAT model at that scale. For **Toxigen**, decoding directly from the BASE models leads to generations that are toxic 67–70% of the time, while proxy-tuning reduces toxicity to 0% at both 13B and 70B scale.

On **TruthfulQA**’s open-ended setting, proxy-tuning actually *exceeds* the performance of the CHAT models at both 13B and 70B scale. Table 3 shows the more granular % Informative and % Truthful scores: proxy-tuning, at 13B and 70B respectively, is 1.0% and 1.4% less informative than the CHAT model, but 3.2% and 6.5% more truthful. The improvement in truthfulness suggests that decoding-time algorithms may provide an avenue for better knowledge preservation, whereas direct tuning has been shown to sometimes hurt performance on knowledge-intensive tasks (Ouyang et al., 2022).

We measure the “gap closed” between each BASE model $\mathcal{M}$ and its directly tuned CHAT version as the difference in performance between $\mathcal{M}$ and the proxy-tuned $\tilde{\mathcal{M}}$, divided by the difference between $\mathcal{M}$ and its CHAT version. On average across the five evaluation settings, proxy-tuning closes 91.1% of the gap at 13B scale, and 88.1% at 70B scale. Moreover, proxy-tuning a larger model outperforms the small expert in all scenarios except for a 0.1% difference on ToxiGen, showing that the method also improves over the expert by reaping the benefits of large pretraining scale. Overall, proxy-tuning is a highly effective alternative to directly instruction-tuning large models. Qualitative examples in Table 1 illustrate that generations from proxy-tuned models are completely fluent and substantially different from those of the base model.

Model	% Info	% True
<i>13B</i>		
Base (untuned)	90.7	56.9
Proxy-tuned	91.4	90.5
Directly tuned	93.0	87.3
<i>70B</i>		
Base (untuned)	93.6	60.0
Proxy-tuned	92.8	92.3
Directly tuned	93.8	85.8

Table 3: More fine-grained results on TruthfulQA.

4 Code Adaptation Experiments

Next we study proxy-tuning on code, due to the availability of downstream tasks and off-the-shelf code models based on LLAMA2. We use CODELLAMA-7B-PYTHON (Rozière et al., 2023) as the expert $\mathcal{M}^+$, which was initialized using LLAMA2-7B, further trained on general code, then specialized on Python code. For readability, we refer to this model as 7B-CODE. Like in §3, we steer 13B- and 70B-BASE as $\mathcal{M}$, and use 7B-BASE as the anti-expert $\mathcal{M}^-$. These experiments test proxy-tuning in a common practical setting where an LM is further pretrained to fit a domain of interest (Gururangan et al., 2020), such as medicine (Wu et al., 2023), scientific text (Beltagy et al., 2019), or non-English languages (Cui et al., 2023).

4.1 Datasets

We evaluate on **CodexEval** (Chen et al., 2021), which asks models to write a Python function given a function signature and description, and **DS-1000** (Lai et al., 2022), which contains Python programming problems from StackOverflow. For both benchmarks, the functional correctness of generated code is automatically evaluated using test cases. We report pass@10, which measures how likely at least one of 10 sampled solutions for a problem is correct, following the setup of Chen et al. (2021). More details in Appendix A.

4.2 Results

Shown in Table 4, proxy-tuning pretrained models on code leads to substantial improvements on coding tasks: at 13B, there is a 32.0% absolute improvement on CodexEval and 16.6% on DS-1000; at 70B, the improvement is 8.6% and 6.7%, respectively.

We observe that in this setting, proxy-tuning a larger model usually does not outperform the tuned 7B-CODE expert alone. Recall that the proxy-tuning equation can be arranged as 7B-CODE + (13B-BASE – 7B-BASE). Because the contrast of (13B-BASE – 7B-BASE) does not improve the 7B-CODE model, we hypothesize that this is because *generic* pretraining at a

Model	CodexEval	DS-1000
<i>7B</i>		
Directly tuned	68.9	53.6
<i>13B</i>		
Base (untuned)	33.7	26.2
Proxy-tuned	65.7	42.8
Directly tuned	78.6	56.9
<i>70B</i>		
Base (untuned)	62.0	43.9
Proxy-tuned	70.7	50.6
Directly-tuned	89.2	67.6

Table 4: **Results for code adaptation.** **Directly tuned** refers to CODELLAMA-PYTHON. The **proxy-tuned** model uses CODELLAMA-7B-PYTHON as the expert, and LLAMA2-7B as the anti-expert. The metric is pass@10 ($\uparrow$).

Model	TriviaQA	GSM
<i>7B</i>		
Directly tuned	55.8	40.6
<i>13B</i>		
Base (untuned)	36.8	6.6
Proxy-tuned	55.9	43.9
Directly tuned	59.5	51.0
<i>70B</i>		
Base (untuned)	45.2	9.6
Proxy-tuned	62.7	53.9
Directly tuned	63.1	67.9

Table 5: **Results for task-specific tuning.** **Directly tuned** refers to a task expert obtained by finetuning LLAMA2 on either TriviaQA or GSM. The **proxy-tuned** model uses the 7B task model as the expert and LLAMA2-7B as the anti-expert.

larger scale is not helpful when the model has already been tuned for a particular domain. This differs from §3, where larger pretraining scale tends to provide more knowledge.

5 Task Finetuning Experiments

Although LMs can respond to arbitrary tasks described in natural language, they usually cannot be reliably applied to specific tasks out-of-the-box. When annotated task demonstrations are available, finetuning is consistently beneficial. Thus, we experiment with proxy-tuning models on particular tasks, including those with specific structural constraints. We consider two tasks: question-answering (TriviaQA) and math word problems (GSM). For each task, we finetune LLAMA2-7B on the train set to obtain a task expert (see §A.3 for details); for comparison, we also tune task experts at 13B and 70B scale. Then, we contrast the task expert and the anti-expert 7B-BASE, to steer 13B- and 70B-BASE.

5.1 Tasks

Question-answering We instantiate the task with trivia questions from **TriviaQA** (Joshi et al., 2017). To obtain task experts, we train models on its 88K train examples to predict the answer given the question. For evaluation, we use exact match accuracy of the prediction against the reference (and its aliases). Exact match is an appropriate metric here as for particular tasks, we usually desire particular answer formats.

Math word problems We use **GSM** (from §3), which contains 7.5K training examples. Given the math question, we train models to predict the answer passage from the dataset. These passages are step-by-step solutions with particular formatting styles, such as enclosing intermediate equations in angle brackets (e.g., “ $\langle 1+1=2 \rangle$ ”) and stating the final answer at the end of the passage following four hash symbols (e.g., “#### 4”).

5.2 Results

As shown in Table 5, proxy-tuning large models with a small, task-specific expert improves performance dramatically. Proxy-tuning 13B-BASE improves absolute performance (over the BASE model alone) by 19.1% on TriviaQA and 37.3% on GSM; for 70B-BASE, the improvement is 17.5% and 44.3%. On average, this closes 84.0% of the gap with the true task expert at 13B, and 86.9% at 70B. Note that the benefit of task adaptation does not decrease as the scale of the base model increases, and proxy-tuning a larger base model (70B compared